

Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering

Sector No. 26, Pradhikaran, Nigdi, Pune 411044

An Autonomous Institute Approved By AICTE and Affiliated To SPPU, Pune

Department Of Information Technology

Software Engineering & Modelling Laboratory
[Course: BIT25PC05]

Mini Project Report

of

T. Y. B. Tech

Academic Year: 2026-27
Semester - I

On

<<MARIS: A Multi-Model AI System for Assessing the Impact of Marine Plastic Pollution on Coral Reefs and Marine Biodiversity>>

By

PRN No.	Name of the Student
124B1F126	**ANANNYA JADHAV**
124B1F128	**ANUSHKA PARKHI**
124B1F129	**VAIDEHI BEHARE**

CERTIFICATE

This is to certify that the project report entitled **MARIS: AI-Based Marine Plastic Pollution Impact Assessment on Biodiversity** has been completed by **Anannya Jadhav, Anushka Parkhi, Vaidehi Behare** under the Project course completion of **Artificial Intelligence and Machine Learning** in Third Year of Information Technology Program. This work is carried out with the entire course requirement as per Artificial Intelligence and Machine Learning concepts in group during Sem- V

Project Guide: Dr. Roshani Raut

ABSTRACT

Plastic waste can now be found from coral reefs to the deep sea yet most reef and seabed monitoring is still carried out by divers or by people who watch hours of underwater video. This project introduces MARIS, a system that uses models to look at underwater pictures and turn them into a measure of ecological pressure. Debris is. Cut out by a YOLO11s-seg network that was trained on TrashCan 1.0. Animals are tracked from one frame to the next with ByteTrack. Coral is divided into bleached and dead parts by a SegFormer (MiT-B2) model that was fine-tuned on the Coralscapes dataset and sea temperature and bleaching warnings are taken from NOAA Coral Reef Watch. These signals, with a water quality index that uses temperature, pH dissolved oxygen, salinity and turbidity are combined by an XGBoost model into a Marine Pollution Impact Score that ranges from 0 to 100. SHAP and Grad-CAM explain each result and all of this is shown in a Streamlit dashboard that helps decision makers. Every model was tested on footage it had never seen before: 53 videos that were kept aside for debris and five dive sites that were unseen for coral. The coral model identified bleaching in 95.7 % of test images separated coral from background with 95.2 % pixel accuracy and achieved an IoU of 0.767. The debris model answered correctly whether plastic was present in 75.1 % of test images compared with a majority baseline of 62.8 % and reached 48.9 % mAP@50 for objects. Fish detection, in deep-sea video remained the weakest part of the work. This work shows that one pipeline can link plastic, fauna, coral and water conditions & that larger and varied labelled data is the main way to improve accuracy.

1. INTRODUCTION

1.1 Background

Every year millions of tonnes of plastic reach the ocean from land [1]. Once there, it drifts, sinks and settles on reefs and on the seabed, where it tangles animals, smothers coral and carries microbes. Field work on Asia-Pacific reefs found that coral touching plastic was far more likely to be diseased [2], while repeated marine heatwaves have already caused mass bleaching across entire reef systems [3]. Managers therefore need to know three things together: how much litter is present, how marine life is behaving around it, and how healthy the coral is.

Today this information is mostly gathered by divers with clipboards or by analysts reviewing underwater video frame by frame. The process is slow, costly and hard to repeat over large areas, and different teams often look at only one of these questions. This project uses the image analysis side of the problem: it builds trained models that read underwater images and video, and connects their outputs to ocean condition data and a single impact estimate shown in a dashboard.

1.2 Importance of AI/ML/Deep Learning

Underwater images are difficult for hand-written rules. Light fades quickly with depth, colours shift towards blue and green, suspended particles blur the view, and a plastic bag can look very similar to a jellyfish or a patch of sand. Deep learning models learn these visual patterns directly from labelled examples instead of relying on fixed thresholds.

Different questions in this project need different kinds of models. Object detection with instance masks (YOLO11-seg [4]) finds each piece of debris and measures its area. Semantic segmentation (SegFormer [5]) labels every pixel of a reef image as live, bleached or dead coral. Multi-object tracking (ByteTrack [15]) links detections of the same animal across frames so that speed and turning can be measured. A gradient boosted tree model (XGBoost [6]) combines these numerical outputs, and explanation methods (SHAP [7] and Grad-CAM [8]) show which inputs and image regions drove a result. Because models can still be wrong, every one of them was evaluated on data kept completely separate from training.

1.3 Problem Definition

The problem addressed is to estimate the ecological pressure caused by underwater plastic from imagery by detecting debris, observing fish movement & assessing coral health & combining these with water conditions into one interpretable score. Existing tools usually solve one of these sub-tasks in isolation, and many published accuracies are measured on images that closely resemble the training images, which overstates real-world performance.

MARIS answers this with a single pipeline: trained detectors and segmenters process each image, satellite and station data supply water conditions, an XGBoost model produces a Marine

Pollution Impact Score (0-100) & a projected biodiversity-loss percentage, and explanations plus recommended actions are shown in a dashboard.

1.4 Objectives

The objectives of the project, taken from the problem brief, are:

1. To detect and classify underwater plastic and other debris in images and video frames.
2. To identify and track fish and other animals and measure behavioural change near plastic.
3. To assess coral reef health and the extent of bleaching.
4. To integrate water quality parameters: temperature, pH, dissolved oxygen, salinity and turbidity.
5. To estimate the impact of plastic pollution on marine biodiversity.
6. To produce a Marine Pollution Impact Score on a 0-100 scale.
7. To provide explainable results and recommendations through a decision-support dashboard.

1.5 Scope

The system works on still images and short video sequences from underwater cameras and remotely operated vehicles. It detects 16 object classes from TrashCan (including plastic, other debris types, the ROV and several animal groups) segments coral into live, bleached and dead regions & reads sea temperature & bleaching alerts for five reference stations from NOAA Coral Reef Watch.

Some parts are outside the current scope. The survey images carry no GPS position, so stations are assigned to them in rotation for demonstration. pH, dissolved oxygen, salinity and turbidity use station reference values rather than live sensors. The impact score and biodiversity-loss figure follow a documented expert weighting so they are indicative estimates rather than measured ecological outcomes. Real deployment would require field validation and more diverse labelled footage.

2. Literature Survey

Research about marine debris, reef images and fish monitoring has increased rapidly thanks, to data sets and new detection tools. Table 2.1 shows the studies and resources that influenced this project and what each one still needs work on.

Table 2.1. Comparative review of related studies and resources

Year [Ref]	Method	Dataset	Reported finding	Limitation / relevance
2019 [9]	YOLOv2, SSD, Faster R-CNN	Trash-ICRA19	Deep detectors can locate underwater litter	Detection only; no link to ecology
2020 [10]	Mask R-CNN baselines	TrashCan 1.0	Instance masks for deep-sea debris and fauna	Benchmark only; our main debris data
2020 [11]	SUIM-Net, U-Net	SUIM	8-class underwater scene segmentation	Reef class has no bleaching label
2024 [12]	Faster R-CNN, YOLOv6	SeaClear	68.3% mAP on random split, ~20% on unseen sites	Shows the cross-site drop we guard against
2025 [13]	SegFormer, DINOv2	Coralscapes	39-class reef segmentation; MiT-B2 mIoU 0.547	Coral only; we connect it to plastic
2022 [14]	Expert-annotated archive	FathomNet	Large labelled deep-sea image database	Used in our fish-detector experiment
2018 [2]	Field statistics	Reef surveys	Plastic contact raises coral disease risk	Manual surveys; motivates automation
2022 [15]	ByteTrack	MOT benchmarks	Keeps low-confidence boxes for tracking	Adopted for fish tracking

The reviewed work shows that deep learning can find debris [9] [10] segment underwater scenes [11] [13]. Follow moving targets [15]. It also shows a recurring weakness: results measured on image splits can fall sharply when the model meets a new site [12]. Each study addresses a task and none connects debris, animal behaviour, coral condition and water quality into one assessment.

2.1 Research Gap and Project Contribution

The gap that exists here is the absence of a combined framework that brings detection, fish behavior, coral health and water quality together to assess the total ecological impact of underwater plastic. MARIS makes a contribution by offering (i) one pipeline and dashboard that covers all four signals, (ii) a model that distinguishes between live, bleached and dead coral, (iii) a leak-free evaluation on videos and dive sites that were not seen before which during the development process revealed and fixed a 77% overlap between training and test data in an earlier data split and (iv) explanations, for each result using SHAP and Grad-CAM. The contribution is the integration and the honest evaluation, not a detection algorithm.

3. Dataset Used

Several public datasets were used, each for the task it suits best. Image datasets. Test the models. Ocean data products supply the environmental condition.

3.1 Dataset Sources

- **TrashCan 1.0 [10]:** underwater ROV video frames of debris and marine animals with instance masks. https://conservancy.umn.edu/handle/11299/214366
- **Coralscapes [13]:** densely labelled Red Sea reef images used for live, bleached and dead coral. https://huggingface.co/datasets/EPFL-ECEO/coralscapes
- **SUIM [11]:** underwater scene segmentation, used for the first U-Net reef model. https://irvlab.cs.umn.edu/resources/suim-dataset
- **FathomNet [14]:** expert-labelled deep-sea fish images used to try to improve fish detection. https://fathomnet.org
- **NOAA Coral Reef Watch [16]:** daily 5 km sea-surface temperature, heat stress and bleaching alerts. https://coralreefwatch.noaa.gov
- **NOAA NCEI Marine Microplastics [17] and OBIS [18]:** measured surface microplastics and species records, used for the plastic map and a real-data check.

3.2 Dataset Description

Table 3.1. Summary of datasets

Dataset	No. of samples	Features	Classes / target	Split used
TrashCan 1.0	7,212 images	RGB pixels, masks, boxes	16 classes (debris, ROV, animals)	4,913 / 986 / 1,313 by video
Coralscapes	2,075 images	RGB pixels, pixel labels	39 classes grouped to 4	1,517 / 166 / 392 by dive site
SUIM	~1,635 images	RGB pixels, pixel labels	8 scene classes	official, 110 test
FathomNet subset	2,720 images	RGB pixels, boxes	fish (4,415 boxes)	training only
NOAA CRW	5 stations, daily	SST, DHW, alert level	water / heat stress	live lookup
NCEI + OBIS	13,460 samples, 698 cells	density, species counts	biodiversity check	cross-validation

3.3 Classes Used

The TrashCan detector predicts sixteen classes. Table 3.2 shows the annotation counts that are most important for this project. Remember, an image can contain objects so the annotation counts refer to objects, not to images.

Table 3.2. Main TrashCan classes and annotation counts

Class	Description	Annotations
trash_plastic	Plastic bags, bottles and fragments	1,864
trash_metal	Cans and metal debris	1,126
trash_etc	Other or unclear debris	2,039
animal_fish	Fish of any species	764
rov	Parts of the recording robot	3,317
Other 11 classes	Paper, fabric, rubber, wood, fishing gear, crabs, eels, starfish, shells, plants, other animals	3,226
Total		**12,336**

For coral the 39 Coralscapes classes were grouped into four targets: live coral, bleached coral and dead coral. Bleached coral is rare so class weights were applied during training.

3.4 Data Preprocessing and Quality

A validation script checked all 7,212 TrashCan images. None were missing or corrupt. No boxes or polygons were invalid. COCO polygon annotations were converted into YOLO segmentation format. Frames from one video look almost identical so the split was made by source video. No video appears in than one of train, validation and test. An earlier split had placed 77% of test images next to copies in training, which inflated mAP to about 0.68. After the fix the honest figure was far lower. All results in this report use the corrected split. Coralscapes was used with its split by dive site so test reefs were never seen in training.

4. Proposed Methodology

MARIS brings together trained models and data sources into one complete assessment. The process includes preparing the data, training the models, evaluating them on held-out data and then putting everything into a dashboard for use.

4.1 Overall Approach

Each image goes through four different analysis branches. First the YOLO11s-seg detector scans the image to find debris, animals and the ROV. Second ByteTrack connects animal detections from one video frame to the next so we can track them over time. Third the SegFormer model. Labels coral pixels in the image. Fourth a water quality module pulls in NOAA temperature data. Combines it with station readings for pH, dissolved oxygen, salinity and turbidity to calculate a Water Quality Index.

All of these outputs are combined into a feature vector. Then an XGBoost model uses this vector to predict the Marine Pollution Impact Score and estimate the projected biodiversity loss. To help understand the results SHAP explains what factors influenced the score and Grad-CAM shows where the detector focused in the image. Finally the dashboard takes all this information. Turns it into clear suggested actions, for users.

4.2 Workflow Diagram

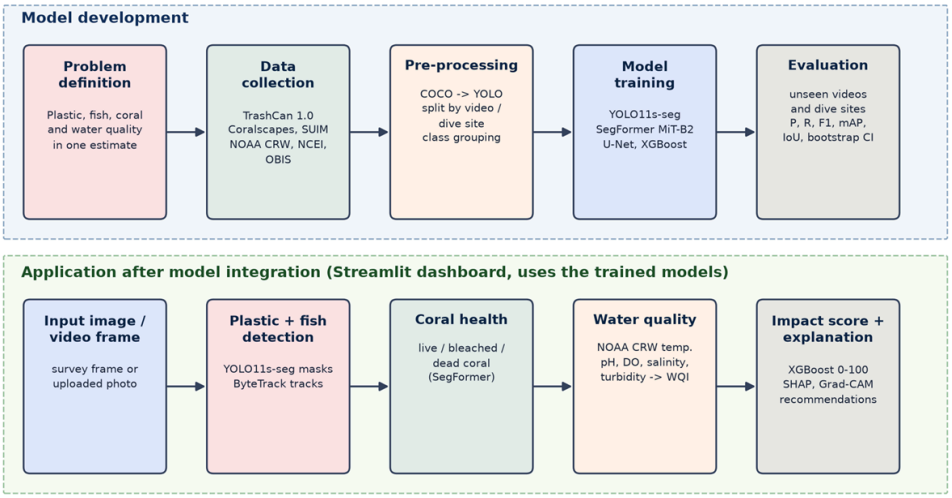


Figure 4.1: Workflow of the proposed MARIS system.

4.3 Preprocessing

TrashCan annotations were converted to YOLO format. Split by video as described in Section 3.4. Training images were resized to 640 pixels with augmentation (mosaic, flips and colour jitter). Coralscapes labels were remapped to four classes and images were processed at 1024 x 2048 resolution with flip test-time augmentation.

4.4 Model Selection

· YOLO11s-seg: single-stage and gives masks so plastic area can be measured and not only counted.

· SegFormer MiT-B2: the strongest model family in the Coralscapes benchmark [13]; a U-Net baseline scored lower.

· ByteTrack: keeps low-confidence detections when linking tracks, which suits faint fish in dark video.

· XGBoost: accurate on small tabular inputs and directly compatible with SHAP explanations.

· SHAP and Grad-CAM: explain the score and the detectors visual attention respectively.

4.5 Implementation Details

The detector was initialised from COCO-pretrained YOLO11s-seg weights. Trained on a Google Colab T4 GPU with automatic mixed precision. The coral model started from the Coralscapes authors released SegFormer weights and was fine-tuned with Dice plus -entropy loss. All thresholds were chosen on validation data only; the plastic confidence threshold is 0.18. The dashboard is written in Streamlit. Runs every model on a selected or uploaded image then passes the result to the impact, map, coral and decision pages.

5. Model Architecture

The central model is YOLO11s-seg. YOLO11s-seg detects and segments the 16 TrashCan classes. YOLO11s-seg outputs feed the rest of the pipeline.

5.1 Architecture Overview

1. Backbone: stacked convolution and C3k2 blocks, plus SPPF pooling and a C2PSA attention block pull out edges, textures and shapes across scales.

2. Neck: upsampling and concatenation using an FPN/PAN style combine coarse features. This lets YOLO11s-seg spot both fragments and big objects.

3. Heads: the detection head outputs boxes and class scores. The segmentation head outputs 32 mask prototypes and per-object coefficients, which together build each instance mask.

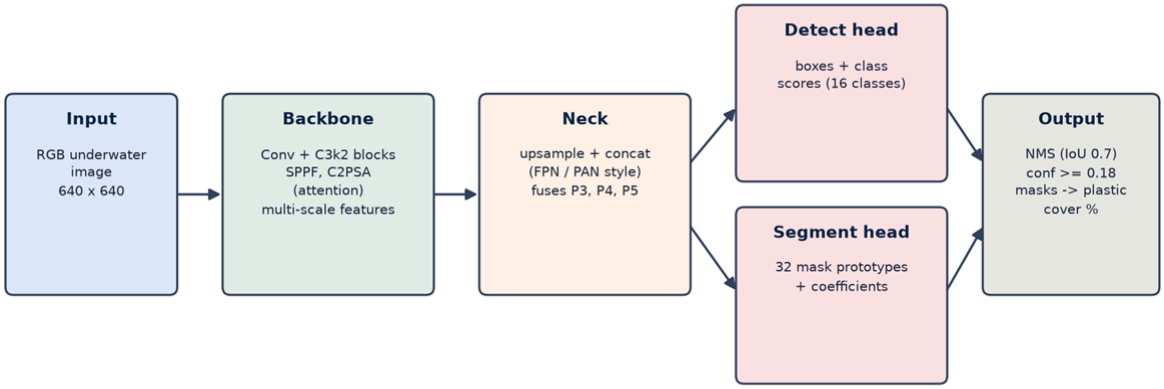


Figure 5.1: YOLO11s-seg architecture as used in MARIS.

5.2 Model Working and Application Integration

An input frame is resized to 640 x 640. The network predicts boxes, classes and confidences along with masks. Non-maximum suppression is used to remove detections. Any detection with a confidence below 0.18 is discarded. Plastic masks are resized back to the image size. The union of these masks gives the percentage of the image covered by plastic. This method ensures that overlapping areas are not counted more than once. At the time the SegFormer model, which has 27.4 million parameters is used to label coral pixels. From this a bleaching severity index is calculated based on the proportion of dead coral. These values along, with the fish behaviour index and the Water Quality Index are then fed into the XGBoost impact model.

6. Mathematical Model

The detectors and segmenters are evaluated with standard overlap and counting metrics, and the impact score is defined as a weighted combination of the four signals.

6.1 Intersection over Union (IoU)

IoU compares a predicted region P with the true region G. A detection counts as correct when IoU >= 0.5.

$$\mathbf{IoU = |P \cap G| / |P \cup G|}$$

6.2 Precision, Recall and F1

With TP true positives, FP false positives and FN missed objects:

$$\mathbf{Precision = TP / (TP + FP) \quad Recall = TP / (TP + FN)}$$

$$\mathbf{F1 = 2 \times Precision \times Recall / (Precision + Recall)}$$

6.3 Mean Average Precision (mAP)

Average Precision (AP) is the area under the precision-recall curve of one class. mAP@50 averages AP over classes at IoU 0.5; mAP@50-95 also averages over IoU thresholds from 0.50 to 0.95.

$$\mathbf{mAP = (1/N) \times \Sigma\, AP_i\,, \; i = 1 \ldots N}$$

6.4 Water Quality Index

Each parameter is scored from 0 to 100 against marine reference ranges, then combined with fixed weights:

$$\mathbf{WQI = 0.30 \cdot S_DO + 0.25 \cdot S_T + 0.20 \cdot S_pH + 0.15 \cdot S_turb + 0.10 \cdot S_sal}$$

6.5 Marine Pollution Impact Score

For reef sites the score combines plastic load P, the fish behavioural anomaly index F, the bleaching severity index B and poor water quality (100 - WQI), each on a 0-100 scale. For sites without coral the coral weight is spread over the other three terms (0.42, 0.29 and 0.29).

$$\mathbf{MPIS = 0.32 \cdot P + 0.22 \cdot F + 0.24 \cdot B + 0.22 \cdot (100 - WQI)}$$

An XGBoost regressor is trained to reproduce this score so that SHAP can attribute every prediction to its inputs.

7. Results and Experiments

This section lists the training setup and the held-out test results. Results are ordered from the module to the weakest.

7.1 Experimental Setup

Debris detection was tested on 1,313 TrashCan images taken from 53 videos that were never used for training, threshold choice or model selection. Coral segmentation was tested on 392 Coralscapes images taken from five dive sites that were absent from training. 95% Confidence intervals for the detector were obtained by bootstrap, over the 53 test videos (1,000 resamples).

7.2 Hyperparameter Details

Table 7.1. Training configuration of the plastic detector

Parameter	Value
Model	YOLO11s-seg
Input image size	640 x 640 pixels
Epochs	120 maximum, early stopping (patience 30)
Batch size	16
Initial learning rate / optimiser	0.01 / auto (AdamW)
Augmentation	Mosaic (off for final 10 epochs), horizontal flip 0.5, HSV jitter
Hardware	Google Colab, NVIDIA Tesla T4 GPU, mixed precision (AMP)
Confidence threshold at inference	0.18 (chosen on validation)

7.3 Evaluation Metrics

- **precision, recall, F1:** share of correct detections, share of objects found, and their balance.
- **mAP@50 and mAP@50-95:** detection quality across confidence levels and box overlaps.
- **IoU and pixel accuracy:** overlap and per-pixel correctness for segmentation.
- **Image-level accuracy:** whether the model correctly says an object type is present in an image, always compared with the majority-class baseline.

7.4 Experimental Results

Table 7.2 shows the model, on unseen dive sites. It was the module. Bleaching was recognised in 95.7% of test images. This is compared to a baseline of 82.9%.

Table 7.2. Coral health results on the Coralscapes test split (392 images, 5 unseen sites)

Metric	Value
Image-level bleaching detection accuracy	**95.7%** (P 84.7%, R 91.0%; baseline 82.9%)**
Image-level coral detection accuracy	**95.4%** (P 97.5%, R 96.9%; baseline 82.7%)**
Coral vs not-coral pixel accuracy	95.2%
Pixel accuracy (4 classes)	94.3%
Coral IoU (precision / recall)	0.767 (0.876 / 0.860)
IoU: live / bleached / dead coral	0.806 / 0.694 / 0.328
Mean IoU (4 classes)	0.693

Table 7.3. Coral model experiments

Experiment	Model	Coral IoU	Mean IoU
1	U-Net (ResNet-34) on SUIM, reef class	0.684	0.684
2	SegFormer v1, ImageNet start, 512 x 1024	0.704	0.647
3	Published Coralscapes SegFormer, no fine-tuning	0.768	0.688
4	**SegFormer v2, fine-tuned, full resolution (ours)**	**0.767**	**0.693**

Table 7.4 asks whether each image contains a given object. ROV detection was strong, debris alarms were correct 89.3% of the time, and plastic presence was judged correctly in 75.1% of images, well above the 62.8% obtained by always answering 'no plastic'.

Table 7.4. Image-level detection on the TrashCan test split (1,313 images, confidence 0.18)

Question	Accuracy	Precision	Recall	Majority baseline
Underwater robot (ROV) present?	**91.5%**	81.5%	90.9%	71.5%
Any debris present?	76.8%	**89.3%**	79.9%	77.9%
Plastic present?	**75.1%**	69.7%	58.4%	62.8%

Object-level plastic detection is harder because each item must be located precisely (Table 7.5). A validation-tuned threshold of 0.47 was also tried; it raised precision but lowered F1, so 0.18 was kept.

Table 7.5. Plastic detector experiments on the TrashCan test split (unseen videos)

Experiment	Threshold	Precision (95% CI)	Recall (95% CI)	F1	mAP@50
1 - default	0.18	47.5% [33.7, 60.2]	45.2% [31.5, 59.4]	**46.3%**	**48.9%**
2 - val-tuned	0.47	61.5% [45.8, 75.8]	33.8% [19.0, 49.6]	43.6%	-

Plastic mAP@50-95 was 34.0%. Among the other classes, the ROV reached F1 61.2%, eels 48.2% and starfish 43.7%. Figure 7.1 shows sample predictions and Figure 7.2 the coral health overlay.

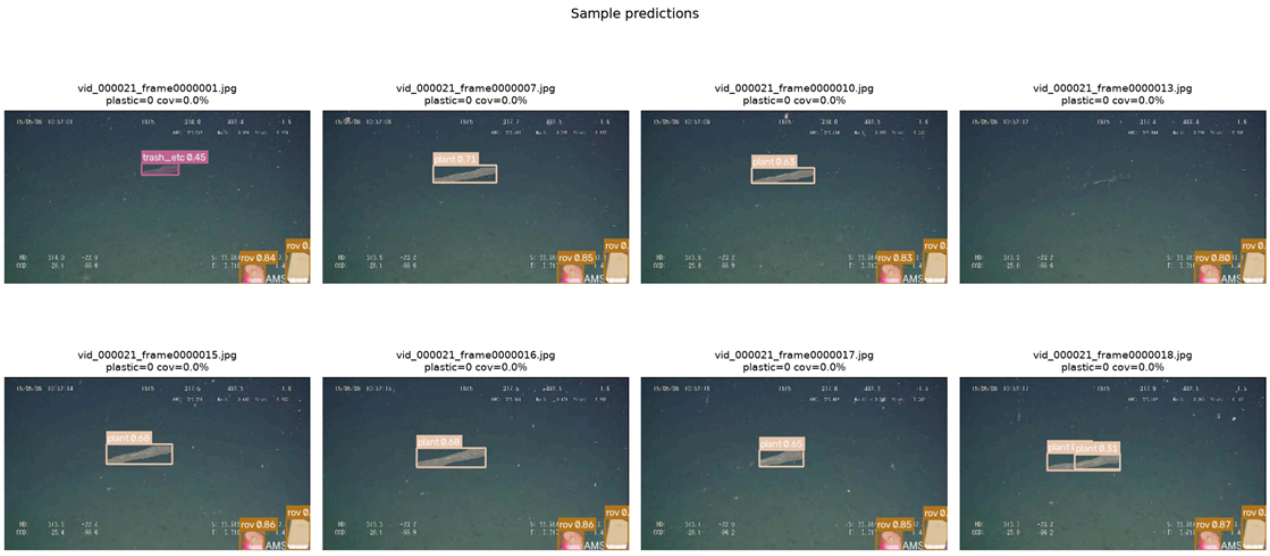


Figure 7.1: Sample detections on held-out TrashCan test frames.

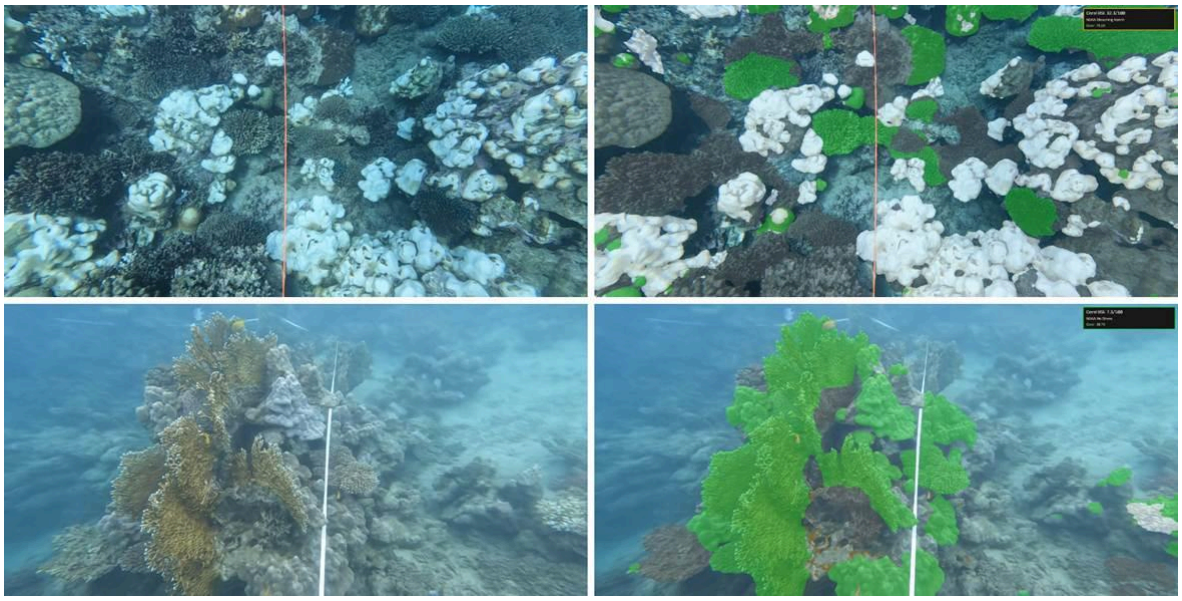


Figure 7.2: Coral health overlay on Coralscapes reef images (left: input, right: model output; green = live, white = bleached coral).

Fish detection was the weakest module (Table 7.6). A second detector trained with extra FathomNet fish images did not improve test F1, and an external fish model that reports 99.3% mAP@50 on its own Brackish data scored only 0.5% AP@50 on DeepFish, which shows how poorly fish detectors transfer to new water.

Table 7.6. Fish detection experiments

Experiment	Test data	Precision	Recall	F1
1 - 16-class detector (ours)	TrashCan test	40.7%	15.9%	22.9%
2 - fish detector + FathomNet (ours)	TrashCan test	26.5%	20.5%	23.1%
3 - Community Fish Detector	OzFish test	76.7%	59.3%	66.9%
4 - Brackish YOLOv8s	DeepFish	4.7%	2.3%	3.1%

Tracking over the 53 test videos produced 34 animal tracks, 8 of them usable moving tracks. Moving animals travelled about 12.3 pixels per frame against 1.75 for near-static ones, but only three tracks passed near plastic, which is too few to draw a behavioural conclusion.

7.5 Fish Detection: Test Details

Table 7.7 gives the full test figures behind Table 7.6. All fish detectors were scored the same way: a predicted box counts as correct when it overlaps a labelled fish with IoU ≥ 0.5 (greedy one-to-one matching), the confidence threshold is chosen on a validation split only, and the test footage comes from videos, camera drops or clips that the threshold choice never saw. The 95% confidence intervals are bootstrapped over those test videos, drops or clips.

Table 7.7. Fish detection test details (IoU ≥ 0.5)

Model	Test data (frames / fish)	Thr.	Precision [95% CI]	Recall [95% CI]	F1	AP@50
16-class detector (ours)	TrashCan, 1,313 / 151	0.18	40.7% [7.1, 62.3]	15.9% [4.2, 27.2]	22.9%	-
Fish detector + FathomNet (ours)	TrashCan, 1,313 / 151	0.18	17.8% [5.9, 29.2]	38.4% [25.3, 52.9]	24.4%	-
Fish detector + FathomNet (ours)	TrashCan, 1,313 / 151	0.35*	26.5% [8.0, 46.4]	20.5% [8.8, 38.7]	23.1%	-
Community Fish Detector (RF-DETR M)	TrashCan, 1,313 / 151	0.85*	12.8% [0.0, 32.8]	7.9% [0.0, 25.4]	9.8%	8.1%
Community Fish Detector (RF-DETR M)	OzFish, 1,144 / 27,802	0.20*	76.7% [75.5, 77.9]	59.3% [56.3, 62.1]	66.9%	65.9%
Brackish YOLOv8s	DeepFish, 758 / 2,767	0.35*	4.7% [1.0, 14.5]	2.3% [0.5, 6.3]	3.1%	0.5%

* threshold chosen on the validation split.

The FathomNet-trained fish detector finds more fish at the default threshold (recall 38.4%) but at the cost of many false alarms, and once its threshold is set on validation data its F1 (23.1%) is no better than the original detector's. Both results lie inside each other's confidence intervals, so the original 16-class detector was kept and is the model that supplies fish detections to tracking and the dashboard.

The pretrained Community Fish Detector works well on shallow, well-lit OzFish footage (66.9% F1 on 27,802 fish) but drops to 9.8% F1 on the dark deep-sea TrashCan frames. At image level it judged whether a fish was present correctly in 86.9% of TrashCan test images, which is below the 92.2% obtained by always answering 'no fish'. The Brackish YOLOv8s model, which reports 99.3% mAP@50 on its own data, found only 65 of 2,767 fish on DeepFish. Inspection of its output showed its 1,312 false alarms falling on coral and along the bottom edge of the frame, while large, clearly visible fish were missed.

Together these tests show that fish detectors do not carry over between water types: each model was strongest on footage like its training data and weak elsewhere. Fish labelled in the target environment are therefore needed. A fine-tuning run on DeepFish, with every training clip that appears in the test set removed, has been prepared but not yet run, so no result is reported for it.

7.6 Regional Environmental Risk Assessment (Gulf of Mannar)

The image pipeline cannot place its survey images on a map, because TrashCan frames carry no GPS position. This module, developed by Anannya Jadhav, works the other way round: it starts from a real location and scores ecological risk across it from measured ocean, species and habitat data. The study area is the Gulf of Mannar and Palk Bay coral region between India and Sri Lanka, bounded by an Allen Coral Atlas polygon. Every ocean grid cell of the Copernicus model inside the polygon is a study site, which gives 245 sites about 9.2 km apart (Table 7.8).

Table 7.8. Data used by the environmental risk module

Source	Variables	Coverage used
Copernicus Marine global analysis (0.083°)	Current speed and direction, sea temperature (daily); significant wave height (3-hourly, averaged per day)	60 days, 4 Aug - 2 Oct 2026; 245 sites
OBIS occurrence records [18]	Species name, class, position, year, depth	Since 2000; 6,208 records, 5,254 after cleaning, 3,536 species
Allen Coral Atlas benthic map	Coral/algae, seagrass, rubble, rock and sand polygons	Whole study polygon

For each site the module computes, within a 10 km radius: species richness, number of taxonomic classes and record density from OBIS; coral cover, seagrass cover, distance to the nearest coral and a habitat sensitivity score from the Coral Atlas (coral 1.0, seagrass 0.8, rubble 0.4, rock 0.3, sand 0.1); and the 60-day mean current speed, wave height and temperature from Copernicus. Across the area the mean current was 0.122 m/s, the mean sea temperature 29.02 °C and the mean wave height 1.28 m. OBIS records were cleaned by keeping species-level names inside the polygon at depths of 200 m or less and removing duplicates.

An XGBoost regressor (200 trees, depth 3, learning rate 0.05) predicts log(1 + species richness) from the seven habitat and ocean features, and SHAP measures how much each feature drives the prediction (Table 7.9, Figure 7.4). The SHAP importances are then used as weights. Habitat features form an ecological value score V and physical features form an

exposure score E, each a weighted sum of features scaled to 0-1. Sites with fewer species than the model predicts receive a biodiversity deficit D (0-1), which raises their score by up to 30%:

$$\text{Impact} = 100 \times \min(1, \sqrt{(V \cdot E)} \times (1 + 0.3 \cdot D))$$

The geometric mean means a site scores high only when it is both valuable and exposed.

Table 7.9. Species richness model and learned weights

Result	Value
XGBoost, 5-fold cross-validation R^2	0.385 (folds 0.18 - 0.60)
XGBoost, nested cross-validation with tuning, R^2	0.325 (folds 0.11 - 0.49)
XGBoost, held-out 20% test R^2	0.157
Random Forest, 5-fold cross-validation R^2	0.427
Value weights (from SHAP)	habitat sensitivity 0.357, seagrass 0.304, coral proximity 0.287, coral cover 0.052
Exposure weights (from SHAP)	wave height 0.420, current speed 0.355, temperature 0.225
Rank stability, 500 random weightings (Spearman)	mean 0.991, 5th percentile 0.968, minimum 0.877

Habitat sensitivity, wave height and seagrass cover had the largest influence on predicted richness. The ranking of sites barely changed when the weights were drawn at random, so the result does not depend on the exact weights chosen. The highest-scoring sites were S005 (8.00° N, 79.75° E, score 67.1), S231 (9.67° N, 79.92° E, 61.5) and S230 (9.67° N, 79.83° E, 57.1): S231 and S230 combine high habitat value with moderate exposure, while S005 is strongly exposed and has far fewer recorded species than its conditions predict (Figure 7.3).

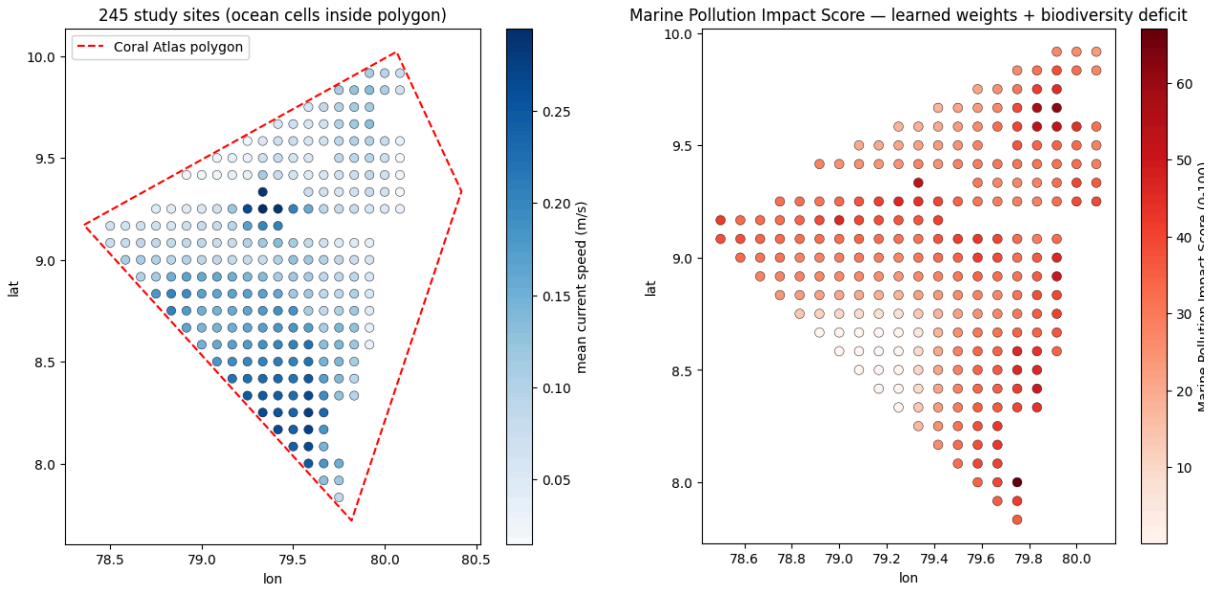


Figure 7.3: Left: the 245 study sites inside the Coral Atlas polygon, coloured by mean current speed. Right: Marine Pollution Impact Score per site.

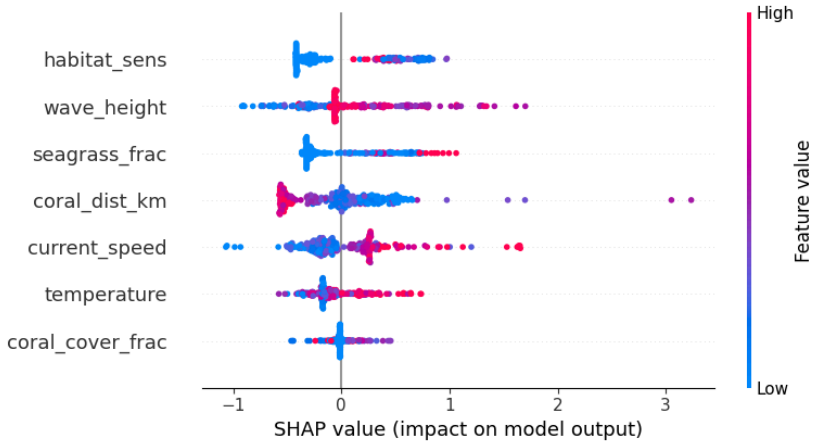


Figure 7.4: SHAP summary of the species richness model.

The module has clear limits. Plastic is not yet part of the score, because no measured plastic data exists for these sites; the score is built so that a plastic term can be added later. The OBIS download carried no IUCN Red List categories, so threatened species could not be weighted. 115 of the 245 sites have no OBIS record within 10 km, so species richness partly reflects where surveys were made. The habitat sensitivity weights are assumptions, and the held-out R^2 of 0.157 shows that the richness model explains only part of the variation; random cross-validation on neighbouring grid cells is also likely to be optimistic. The site scores therefore rank relative risk within the study area and are not measured biodiversity loss. The model results are shown on the Biodiversity Loss Prediction page of the dashboard and the site map on the Marine Pollution Impact Score page.

8. Results Discussion

The results show a ranking. Coral health can be assessed reliably from images. On reefs the model, which had never seen bleaching before identified bleaching in 95.7 percent of images and achieved an Intersection over Union (IoU) of 0.767. Dead coral, with an IoU of 0.328 was the coral class to detect because it resembles rubble and rock.

Large distinctive objects are detected well. The ROV was recognised in 91.5 percent of images and debris alarms were usually correct. Small plastic items are harder to find giving an object-level F1 score of 46.3 percent. This is a figure for unseen videos. The SeaClear authors report a drop from 68 percent to 20 percent mean Average Precision when testing on new sites. Error analysis shows that plastic, metal and fishing gear suffer from both misses and false alarms whereas fish and paper are mostly missed. Rare classes such as shells, fabric and rubber had few training examples.

Fish detection in dark deep-sea footage remains the weakness so the fish behaviour analysis is currently descriptive. The impact score and biodiversity-loss estimate follow expert weighting.

ANNEXURE

Annexure A: Code Repository

The source code is kept in the project Git repository. It contains the data preparation, training and evaluation scripts, the Colab notebooks, the analysis modules and the Streamlit dashboard.

GitHub Repository: https://github.com/wizboltez/MARIS

- src/modules/ - plastic quantification, fish tracking, coral health, water quality, impact scoring, SHAP and Grad-CAM
- src/evaluation/ - held-out evaluation scripts with bootstrap confidence intervals
- notebooks/ - Colab training notebooks for the detector, coral and fish models
- app.py - Streamlit decision-support dashboard (run: streamlit run app.py)
- outputs/metrics/ - every reported number as JSON / CSV

Annexure B: Additional Results

Supplementary graphs produced during training and evaluation.

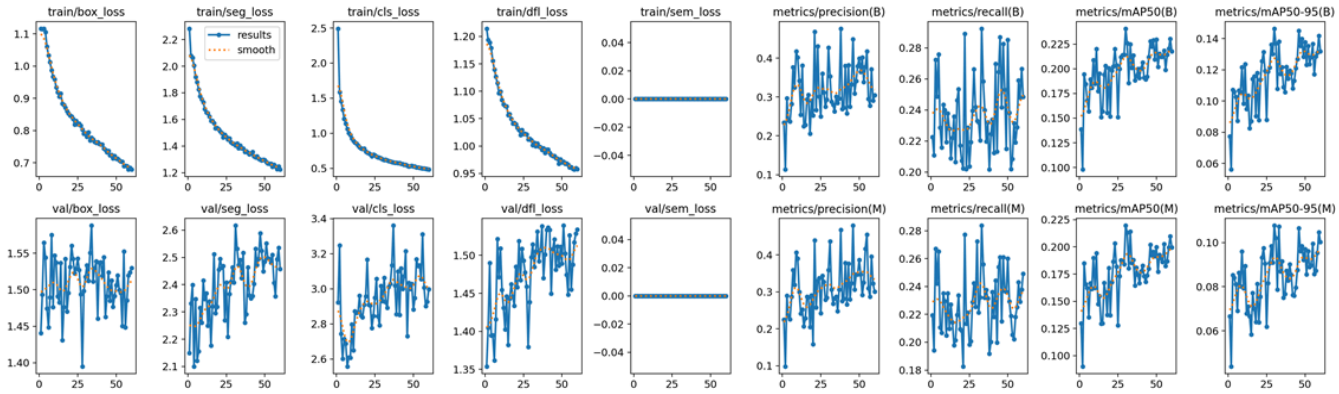


Figure B.1: Training and validation curves of the YOLO11s-seg detector.

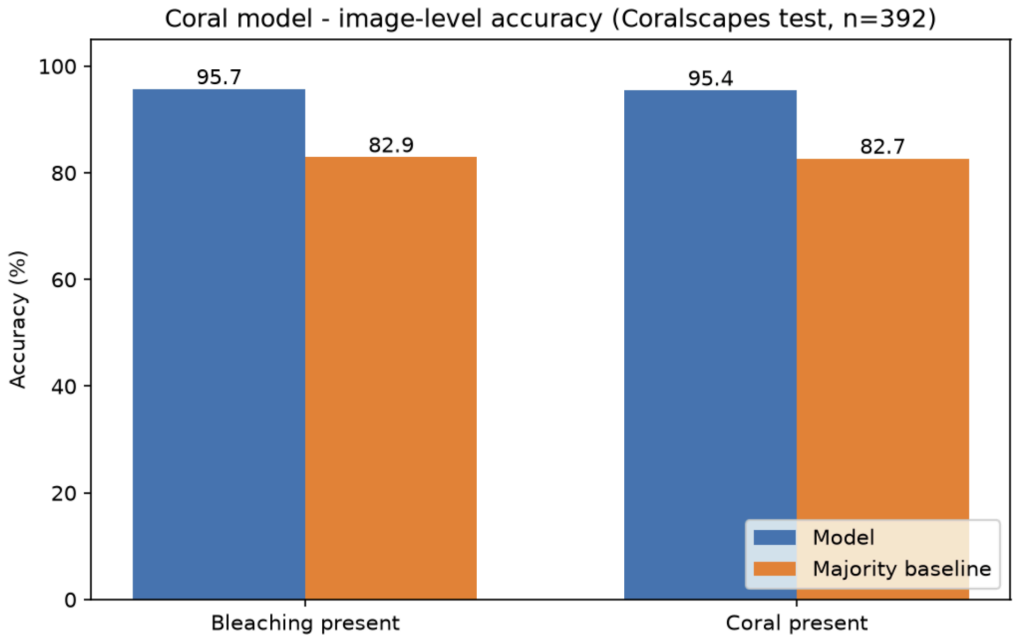


Figure B.2: Coral model accuracy compared with the majority baseline on the Coralscapes test set.

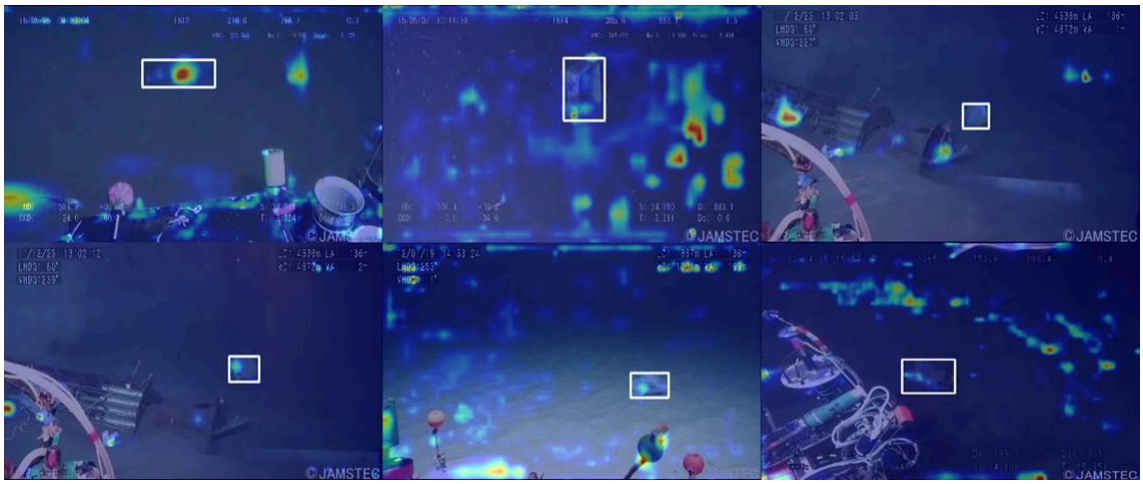


Figure B.3: Grad-CAM views of where the detector looks.

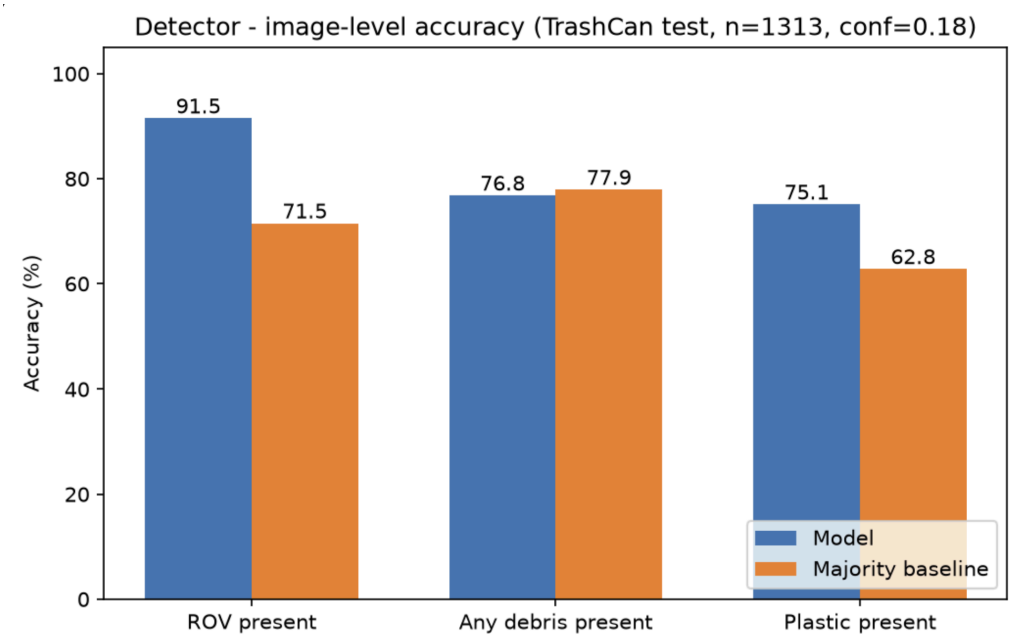


Figure B.4: Image-level detection accuracy compared with the majority baseline on the TrashCan test set.

Annexure C: Implementation Evidence

Screenshots and outputs from the MARIS dashboard covering the six expected outcomes of the brief.

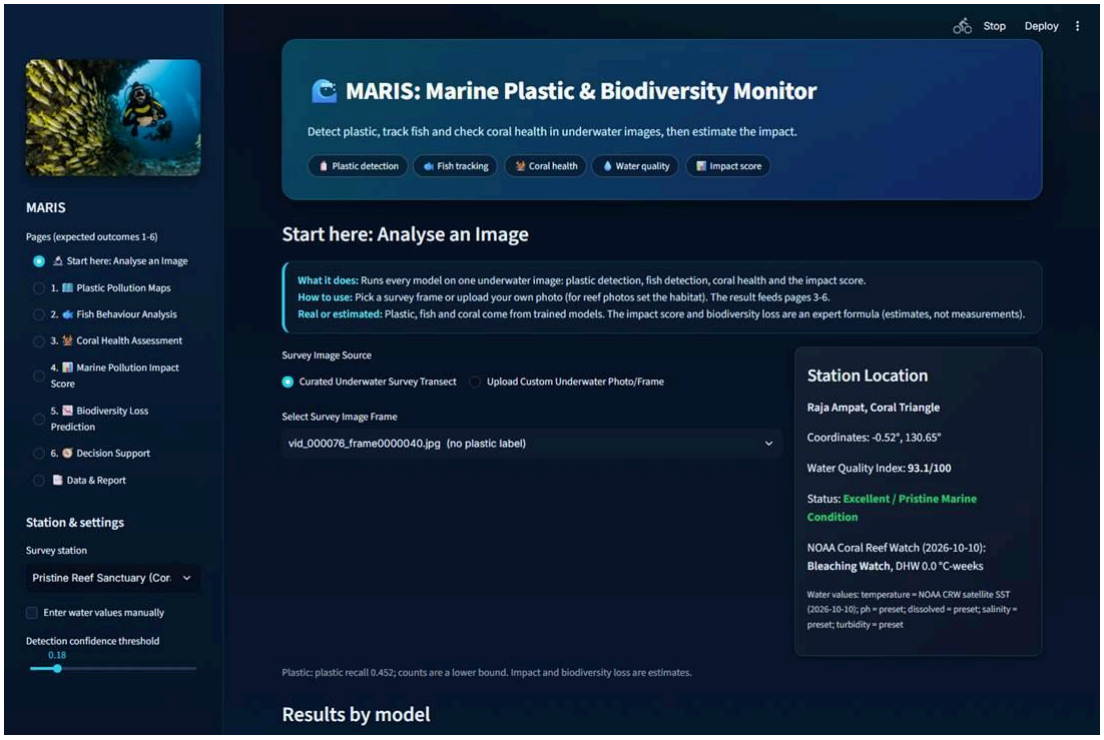


Figure C.1: MARIS dashboard - image analysis page with the six outcome pages.

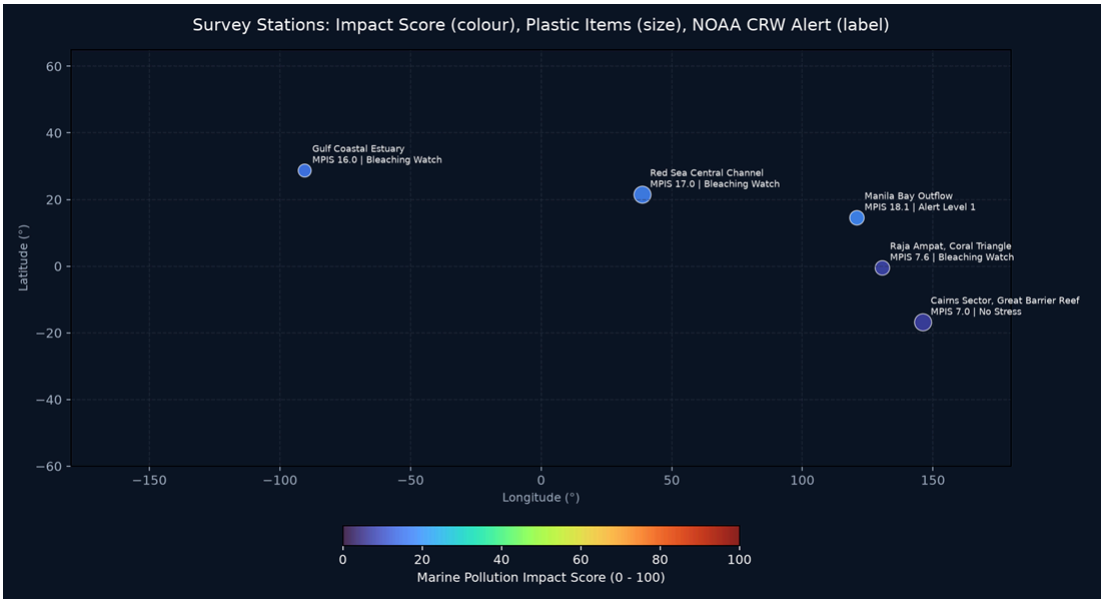


Figure C.2: Plastic pollution map from NOAA NCEI measurements.

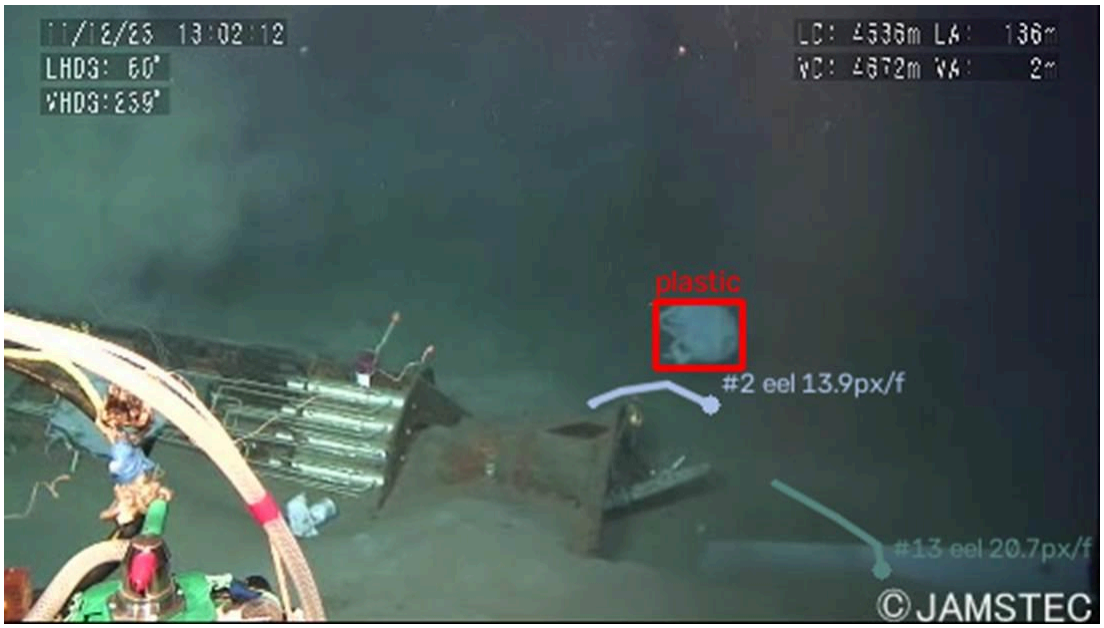


Figure C.3: Fish tracks and behaviour near plastic.

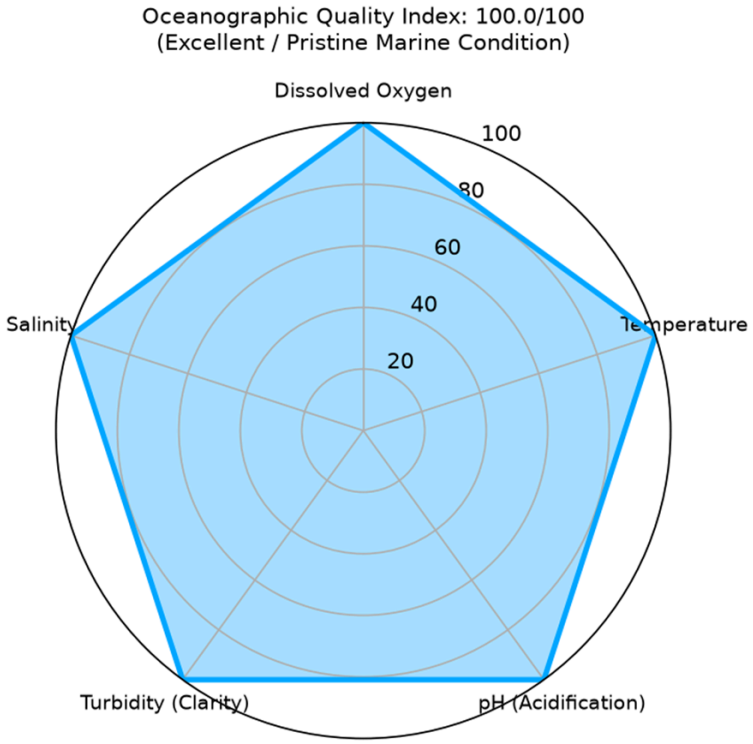


Figure C.4: Water quality parameters against reference ranges.

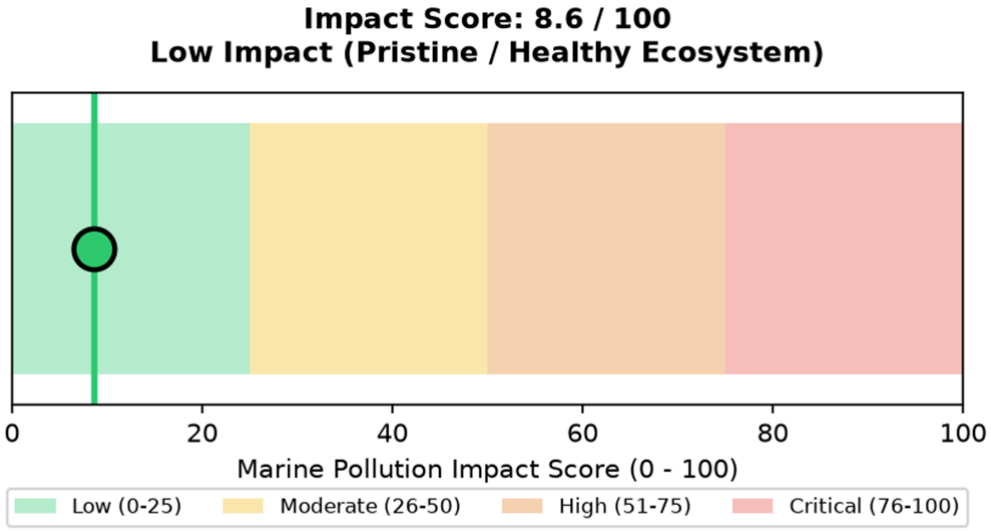


Figure C.5: Marine Pollution Impact Score for one survey image (8.6 / 100, Low impact band).

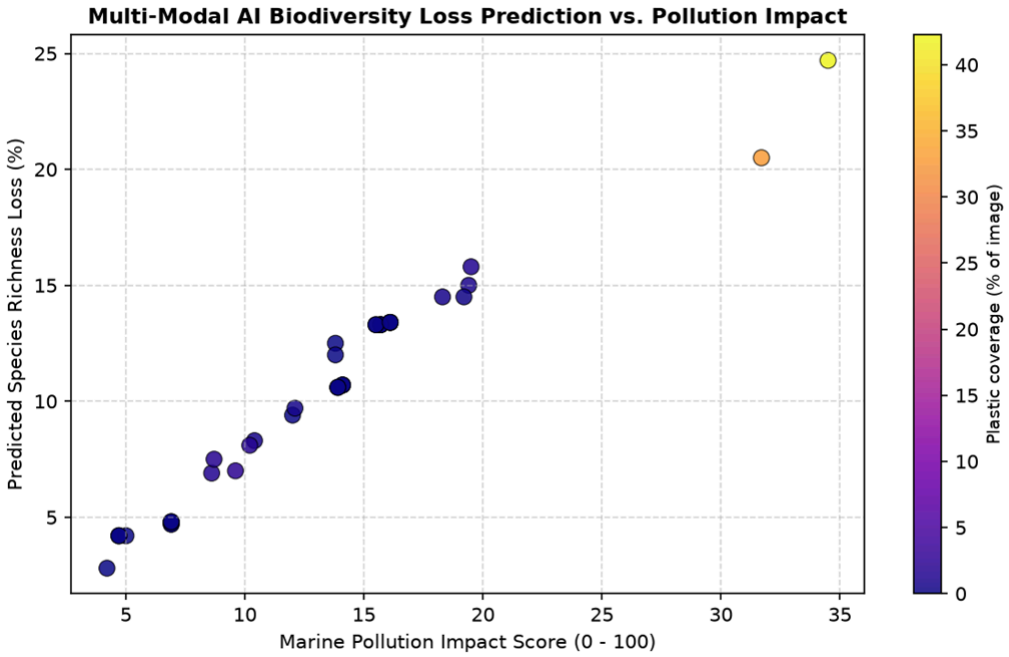


Figure C.6: Projected biodiversity loss vs. impact score .

Annexure D: Team Contributions

Contributions of each team member to the MARIS project.

Table D.1. Team contributions

No.	Team member	Contribution
1	Anannya Jadhav	[describe contribution]
2	Anushka Parkhi	[describe contribution]
3	Vaidehi Behare	[describe contribution]